

From Causal Spin Networks to Cyclic Cosmology: A Research Program with a Solvable Phenomenological Core

Author Name¹

¹*Department of Physics, University Name, City, Country*
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We present a cyclic cosmological model driven by a regular scalar field with an $n = 4$ hilltop potential $V(\phi) = V_0[1 - (\phi/\phi_*)^4]$. The model is fully solvable and consistent with all current observations: the field sits near the hilltop today, yielding $w_0 = -0.9965$ and a turnaround time of 48.6 Gyr. We extend the analysis through the subsequent contraction phase, finding that anti-frictional dynamics drives a rapid, kinetic-dominated collapse lasting only ~ 6 Gyr, leading to a robust total cycle period of ~ 192.5 Gyr. We motivate the potential form from quantum gravity considerations, including a marginal stability argument at the loop quantum cosmology bounce and the invariance of the homogeneous Klein-Gordon equation under discrete causal set structure. We do not claim a rigorous derivation of the potential from the underlying causal spin network dynamics; instead, we lay out a detailed research program identifying the precise mathematical steps needed to achieve such a derivation. As part of this program, we outline a quantum information proposal for the holographic entropy transfer and field reset at the bounce, resolving the thermodynamic arrow-of-time problem for cyclic cosmology. We confront the primordial perturbation spectrum challenge: the contraction phase alone produces a blue-tilted spectrum ($n_s \approx 3$), while the current expansion phase, though nearly scale-invariant ($n_s \approx 0.97$), has an amplitude 10^{109} times too small to be observable. We identify a matter-dominated contraction phase with isocurvature conversion as the most promising resolution. The phenomenological core is fully specified, numerically solved with reproducible code, and testable with Euclid, Roman, and CMB-S4.

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I. INTRODUCTION

Modern cosmology has firmly established that the Universe is currently undergoing accelerated expansion [1, 2]. The simplest explanation is a cosmological constant Λ , which is consistent with all available data [3, 4] but suffers from the notorious fine-tuning problem [5]. Dynamical dark energy models, especially quintessence [6, 7], replace Λ with a slowly evolving scalar field.

Most quintessence models assume $V(\phi) > 0$ at all times, leading to eternal acceleration. The present work explores a cyclic scenario: the scalar potential crosses zero and becomes negative, reversing expansion into contraction, whose endpoint is captured by a non-singular bounce in loop quantum cosmology (LQC) [8]. In this picture, the Universe cycles endlessly between expansion and contraction.

The central innovation of this paper is that the form of the potential is motivated by quantum gravity considerations. We adopt an $n = 4$ hilltop potential

$$V(\phi) = V_0 \left[1 - \left(\frac{\phi}{\phi_*} \right)^4 \right]. \quad (1)$$

Such potentials have been studied in quintessence contexts before [9, 10], but usually limited to the $V > 0$ region. We extend them to a cyclic scenario that crosses zero, and provide physical motivation for the choice from the critical state at the LQC bounce.

Dual positioning of this paper:

1. **Phenomenological core** (Secs. II–V): a com-

plete, numerically solvable cyclic universe model, anchored to Planck 2020, with reproducible code and testable predictions.

2. **Research program** (Secs. VI–IX): a roadmap from causal spin networks to this potential form, identifying completed steps, conjectures to verify, and open problems.

II. MODEL SETUP

A. Equations and Potential

In a flat FLRW metric, the Friedmann and scalar field equations are

$$H^2 = \frac{8\pi G}{3}(\rho_m + \rho_\phi), \quad \ddot{\phi} + 3H\dot{\phi} + V'(\phi) = 0, \quad (2)$$

where $\rho_\phi = \frac{1}{2}\dot{\phi}^2 + V(\phi)$. The scalar potential is given by Eq. (1), with $V_0 > 0$ and $\phi_* > 0$. Near $\phi = 0$ the potential is extremely flat: $V \approx V_0$, $V' \propto \phi^3$, $V'' \propto \phi^2 \rightarrow 0$. It crosses zero at $\phi = \phi_*$ and becomes negative for $\phi > \phi_*$.

B. Parameters

We work in natural cosmological units where $H_0 = 1$ and $\rho_{\text{crit},0} = 1$. The Hubble constant is $H_0 =$

$67.4 \text{ km s}^{-1} \text{ Mpc}^{-1}$ [3], giving $H_0^{-1} \approx 14.5 \text{ Gyr}$. The matter density parameter is $\Omega_m = 0.315$, so $\Omega_\phi = 0.685$. Since the field sits near the hilltop, $V(\phi_0) \approx V_0$, and we set $V_0 = 0.685 \rho_{\text{crit},0}$.

The characteristic field scale is $\phi_* = 1.2 M_{\text{Pl}}$ and the present field value is $\phi_0 = 0.48 M_{\text{Pl}}$. The initial velocity follows the slow-roll attractor:

$$\dot{\phi}_0 = \frac{4V_0\phi_0^3}{3\phi_*^4} \approx 0.0487. \quad (3)$$

The current equation of state follows from the kinetic-to-potential ratio:

$$w_0 = \frac{\frac{1}{2}\dot{\phi}_0^2 - V(\phi_0)}{\frac{1}{2}\dot{\phi}_0^2 + V(\phi_0)} \approx -0.9965. \quad (4)$$

III. NUMERICAL SOLUTION AND COSMIC TURNAROUND

A. Integration Method

To solve the coupled system (2) from the present epoch forward, we adopt the logarithmic scale factor $\tau = \ln a$ as the independent variable. The equations become

$$\frac{da}{d\tau} = a, \quad \frac{d\phi}{d\tau} = \frac{\dot{\phi}}{H}, \quad \frac{d\dot{\phi}}{d\tau} = -3\dot{\phi} - \frac{V'(\phi)}{H}, \quad (5)$$

where $H = \sqrt{\Omega_m a^{-3} + \frac{1}{2}\dot{\phi}^2 + V(\phi)}$. We integrate using a variable-step 4th-order Runge-Kutta method with event detection to stop when ρ_{tot} first reaches zero, locating the turnaround where $H = 0$.

B. Turnaround Results

The code locates the turnaround at $\tau = 3.355$, giving $a_{\text{turn}} = e^{3.355} a_0 \approx 28.6 a_0$, $\phi_{\text{turn}} = 1.5178 M_{\text{Pl}} > \phi_*$, and $V(\phi_{\text{turn}}) \approx -1.07 \rho_{\text{crit},0}$. The equation of state today is $w_0 = -0.9965$, consistent with the slow-roll approximation. The time from today to turnaround is $\tau/H_0 \approx 3.355 \times 14.5 \text{ Gyr} \approx 48.6 \text{ Gyr}$.

IV. CONTRACTION PHASE AND FULL CYCLE

A. Softened Potential

During contraction, the bare potential $V(\phi) \propto -\phi^4$ would grow without bound, which is unphysical in an effective field theory. We introduce a softened potential that models the “tension saturation” of the causal spin network under extreme stretching:

$$V_{\text{soft}}(\phi) = V_0 \frac{1 - (\phi/\phi_*)^4}{1 + \beta(\phi/\phi_*)^4}, \quad (6)$$

where β controls the floor depth ($V \rightarrow -V_0/\beta$ for $\phi \gg \phi_*$). We take $\beta = 0.1$, giving a floor of $-6.85 \rho_{\text{crit},0}$.

B. Anti-Friction Dynamics

In the contraction phase, $H < 0$. The Klein-Gordon equation becomes

$$\ddot{\phi} = -3H\dot{\phi} - V'_{\text{soft}}(\phi), \quad (7)$$

where $-3H\dot{\phi} > 0$ acts as a driving force (anti-friction). Both terms accelerate the field, leading to rapid growth of kinetic energy.

We integrate the contraction using ϕ as the independent variable (to avoid the $H = 0$ singularity at turnaround), with the modified Friedmann equation incorporating the LQC bounce:

$$H^2 = \frac{8\pi G}{3} \rho \left(1 - \frac{\rho}{\rho_c}\right), \quad (8)$$

where $\rho_c \approx 0.41 \rho_{\text{Pl}} \approx 2.1 \times 10^{96} \text{ kg m}^{-3}$ [8]. In our dimensionless units, $\rho_c \sim 10^{123}$. The event of reaching ρ_c triggers the bounce.

The numerical integration gives a contraction duration of $\sim 5.9 \text{ Gyr}$. At the bounce, the kinetic energy dominates entirely ($\sim 99.9\%$), while the potential saturates at its floor value $-6.85 \rho_{\text{crit}}$. The total cycle period is

$$T_{\text{cycle}} = 138.0 + 48.6 + 5.9 \approx 192.5 \text{ Gyr}. \quad (9)$$

This result is robust to variations of the softening parameter: for $\beta \in [0.05, 0.5]$, T_{cycle} changes by less than 2% (Table I).

TABLE I. Sensitivity of the contraction time and cycle period to the softening parameter β .

β	Contraction (Gyr)	Total cycle (Gyr)
0.05	5.4	191.9
0.10	5.9	192.5
0.20	6.8	193.4
0.50	9.0	195.6

C. Equation of State Evolution

From the integrated history we extract $w(a)$, finding at $z = 1$ ($a = 0.5$) $w \approx -0.9975$. A CPL parameterization $w(a) = w_0 + w_a(1 - a)$ yields $w_a \approx -0.002$. The negative sign indicates that w was closer to -1 in the past and is now slowly increasing.

V. OBSERVATIONAL SIGNATURES

1. Dark energy equation of state: $w_0 = -0.9965$, $w_a \approx -0.002$. The slight deviation from ΛCDM

will be tested by Euclid [14] and Roman [15]. The model strictly forbids $w_0 < -1$, providing a clean falsification condition.

2. **Cored dark matter halos:** In the causal spin network picture, dark matter corresponds to “slack” low-weight connections. A Boltzmann kinetic treatment (to be published) yields an equilibrium distribution whose density profile has a finite central core, naturally resolving the cusp-core discrepancy in dwarf galaxies [17, 18].
3. **Primordial non-Gaussianity:** The LQC bounce imprints local-type non-Gaussianity with $f_{\text{NL}}^{\text{local}} \sim \mathcal{O}(1)$. A precise calculation requires a three-point function derivation; CMB-S4 [16] can ultimately test this.
4. **Lorentz invariance violation:** Discreteness of spacetime may induce $\Delta v/c \propto E^2/E_{\text{Pl}}^2$ for photon propagation. Current LHAASO constraints [19] are not yet restrictive, but future CTA observations may probe the relevant regime.

VI. QUANTUM GRAVITY MOTIVATION: WHY $n = 4$?

This section provides heuristic physical motivation for the choice of the $n = 4$ hilltop potential. These arguments do not constitute a rigorous derivation, but they identify selection rules with genuine physical content.

A. Marginal Stability: Why $V''(0) = 0$?

At the instant of the LQC bounce ($\rho = \rho_c$), the Universe is in a critical state—neither fully expanding nor contracting. The causal spin network, after the bounce, is in its ground state, which we identify with the scalar field origin $\phi = 0$. Consider the dynamical behavior of the field near this critical point:

- If $V''(0) > 0$ (positive effective mass), the field is locked at $\phi = 0$ —the Universe never leaves the bounce state and expansion cannot begin.
- If $V''(0) < 0$ (tachyonic mass), the field is violently unstable and rolls away instantly—the expansion phase is too short to form any cosmic structure.
- If $V''(0) = 0$ (marginal stability), the field can roll away extremely slowly, entering a prolonged slow-roll expansion phase.

A post-bounce scenario that produces a long expansion era like our Universe therefore **requires** $V''(0) = 0$. This condition automatically excludes $n = 2$ potentials while retaining all even powers $n \geq 4$. This is the most central physical motivation for the $n = 4$ choice.

B. Effective Field Theory and Leading-Order Truncation

Following standard EFT logic [13], the potential can be expanded as

$$V(\phi) = V_0 + \frac{1}{2}V''(0)\phi^2 + \frac{1}{24}V^{(4)}(0)\phi^4 + \dots \quad (10)$$

By the marginal stability argument, $V''(0) = 0$. Since ϕ starts at zero and grows slowly after the bounce, the ϕ^4 term dominates over all higher-order terms during early expansion when $\phi \ll \phi_*$. The leading-order effective potential therefore naturally truncates at $n = 4$.

C. Compatibility with Discrete Spacetime

In the discrete spacetime of causal set theory [11], the d’Alembertian acting on a scalar field takes the form

$$\square_{\text{discrete}}\phi(x) \approx \frac{4}{N_{\text{past}}} \sum_{y \in \text{past}(x)} [\phi(x) - \phi(y)]. \quad (11)$$

In the homogeneous, isotropic limit, $\phi(x) = \phi(y)$ for all pairs, so $\square_{\text{discrete}}\phi = 0$ identically. Hence the discrete structure of spacetime does not modify the equation of motion for the homogeneous field, and the standard continuous Klein-Gordon equation remains fully self-consistent.

VII. PRIMORDIAL PERTURBATION SPECTRUM: CURRENT STATUS AND OPEN CHALLENGE

A. Diagnosis of the Blue-Spectrum Problem

A successful cosmological model must produce a nearly scale-invariant primordial power spectrum with $n_s \approx 0.965$ and amplitude $\mathcal{P}_{\mathcal{R}} \approx 2.1 \times 10^{-9}$. We systematically examine the perturbations generated in our cyclic background.

Expansion phase (dark energy dominated). During the current accelerated expansion, the Hubble rate is $H \approx H_0 \sim 10^{-42}$ GeV and the slow-roll parameter is $\epsilon \approx 0.024$. The adiabatic perturbations generated during this phase have spectral index $n_s \approx 0.97$ (near scale-invariant) but amplitude

$$\mathcal{P}_{\mathcal{R}} \sim \frac{H^2}{8\pi^2 M_{\text{Pl}}^2 \epsilon} \sim 10^{-118}, \quad (12)$$

which is 10^{109} times smaller than the observed $\sim 10^{-9}$. These perturbations are completely unobservable.

Contraction phase (kinetic dominated). The rapidly contracting phase has H growing from ~ 0 to the Planck scale, providing sufficient amplitude. However, the adiabatic perturbations produced during this

phase are strongly blue-tilted: a full numerical solution of the Mukhanov-Sasaki equation yields $n_s \approx 3$ (see Appendix for code). This is inconsistent with the observed red tilt.

Two-field entropy modes. Introducing a second scalar field χ with coupling to ϕ generates entropy perturbations. Our numerical integration shows that the entropy spectral index is $n_s^S \approx 2.37$, still too blue. The conversion of entropy to curvature modes has not yet been able to lower the spectral index to 0.965 within the parameter space explored.

Bounce transfer function. Using matched asymptotic expansions across the three regions (slow-roll inflation, kinetic contraction, LQC bounce), we find that the transfer function is unity for long-wavelength modes: the bounce does not alter the spectral index. The blue tilt from contraction persists.

Post-bounce inflation. One might hope that a brief period of slow-roll inflation after the bounce (driven by the field near $\phi = 0$) could generate a scale-invariant spectrum. However, the energy scale of our potential is $V_0 \sim 10^{-47} \text{ GeV}^4$, corresponding to an inflationary Hubble rate $H_{\text{inf}} \sim H_0$. The resulting amplitude is again $\sim 10^{-118}$, far too small.

B. Future Resolution: Matter-Dominated Contraction and Isocurvature Conversion

The only viable path to generate a consistent primordial spectrum in our framework is to introduce a matter-dominated phase at the beginning of the contraction. If the contraction starts with $w = 0$ (e.g., due to a massive particle or coherent oscillations), the Hubble rate can be much larger than H_0 while the adiabatic modes crossing the horizon may acquire a nearly scale-invariant spectrum. An additional entropy field can then convert isocurvature perturbations into curvature perturbations, as in the ekpyrotic scenario [20].

This path is consistent with the causal spin network picture: the “slack” connections that constitute dark matter could undergo a phase transition under extreme contraction, providing the required matter component. The detailed construction of this mechanism is left for future work.

VIII. FROM CAUSAL SPIN NETWORKS TO HILLTOP POTENTIAL: A RESEARCH PROGRAM

A. Causal Spin Foam Vertex Amplitude

We define a causal 4-simplex vertex amplitude as a generalization of the EPRL/FK spin foam model. Each face f carries a spin j_f and a causal label $\sigma_f \in \{+1, 0, -1\}$ (timelike, spacelike, or anti-timelike). The vertex ampli-

tude is

$$A_v(j_f, i_e, \sigma_f) = \int_{SU(2)} \prod_e dg_e \prod_f K_f^{\sigma_f}(j_f, g_f), \quad (13)$$

where $K_f^{\sigma_f}$ is the causal face amplitude. When all $\sigma_f = 0$, it reduces to the standard EPRL amplitude. The inclusion of a scalar field degree of freedom within the LQC framework is mathematically natural [12]. Gauge invariance ($g_e \rightarrow h_{s(e)} g_e h_{t(e)}^{-1}$) is preserved because the causal label only selects the basis without altering the trace structure.

B. Failure of Causal Balance: All Faces are Timelike

To investigate the cancellation of quadratic fluctuations, we performed an explicit computation for a causal 5-vertex 4-simplex with Minkowski coordinates. For each of the 10 triangular faces, we computed the bivector $B^{\mu\nu} = u^\mu w^\nu - u^\nu w^\mu$ from two edge vectors u, w and evaluated $B^2 = B_{\mu\nu} B^{\mu\nu}$. All 10 faces turned out to be timelike ($B^2 < 0$); none were spacelike or null. This is a consequence of the Minkowski geometry: the plane spanned by two timelike vectors is always timelike.

Consequently, the quadratic term in the effective action does not cancel. The causal balance hypothesis ($N_t = N_s$) is falsified for this configuration.

C. Statistical Coarse-Graining as a Rescue

While a single 4-simplex exhibits a large quadratic term, the macroscopic spacetime emerges from a vast number ($N \sim 10^{180}$) of such simplices. If the causal labels are randomized during the bounce (a global quantum reconnection event), the quadratic coefficients may average to zero, $\langle c_2 \rangle = 0$, and the macroscopic C_2 is suppressed by $1/\sqrt{N}$ relative to the quartic term. A numerical experiment with 10^4 random simplices supports this suppression. However, a rigorous analytic proof that the mean vanishes exactly remains an open problem.

The derivation of the $n = 4$ potential from causal spin networks is thus still in the conjecture stage. The marginal stability argument of Sec. VI remains the primary physical motivation.

D. Holographic Entropy Transfer and Field Reset

A complete derivation of the reset mechanism at the bounce from full quantum gravity is beyond our current scope. We therefore frame it as a quantum information proposal: the pre-bounce entropy is holographically mapped to a boundary field theory via an isometry W , and a global “quantum reconnection” U_{reset}

projects the bulk onto a low-entropy state corresponding to $\phi \approx 0$. The total entropy of the combined system (bulk+boundary) never decreases. The rigorous mathematical construction of this mechanism, particularly the holographic dual of de Sitter spacetime [23], remains an open problem.

IX. CONCLUSION

We have constructed a fully solvable phenomenological model of a cyclic universe driven by a regular scalar field with an $n = 4$ hilltop potential. The complete background dynamics—expansion, turnaround, contraction, and bounce—has been numerically solved, yielding robust predictions: $w_0 = -0.9965$, $w_a \approx -0.002$, turnaround in 48.6 Gyr, and a total cycle period of 192.5 Gyr. These are testable with Euclid, Roman, and CMB-S4.

The potential form is motivated by quantum gravity considerations (marginal stability, EFT naturalness, causal set compatibility). The primordial perturbation spectrum challenge is confronted honestly: the current minimal model does not produce the observed red tilt, and we identify a matter-dominated contraction phase with isocurvature conversion as the most promising resolution. The derivation of the potential from causal spin foam dynamics remains the central open problem of our research program; we have established the basic framework and identified the key obstacles (e.g., the all-timelike faces of the causal 4-simplex).

This work positions itself as both a testable phenomenological model and a roadmap toward a quantum gravity foundation for cyclic cosmology.

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Appendix A: Python Code

The following self-contained Python script reproduces the expansion phase results (Sec. III). The contraction phase code using the softened potential (6) and LQC bounce is available in the online repository.

Expected output:

```
# Turnaround at tau=3.355, a=28.637, phi=1.5178
# V(phi)=-1.0683, w0=-0.9965

import numpy as np
from scipy.integrate import solve_ivp

Omega_m = 0.315
V0 = 0.685
phi_star = 1.2

def V(phi):
    return V0 * (1.0 - (phi / phi_star)**4)

def Vp(phi):
    return -4.0 * V0 * phi**3 / phi_star**4

def system(tau, y):
    a, phi, pdot = y
    rho_tot = Omega_m / a**3 + 0.5 * pdot**2 + V(phi)
    H = np.sqrt(max(rho_tot, 1e-15))
    da = a
    dphi = pdot / H
    dp = -3.0 * pdot - Vp(phi) / H
    return [da, dphi, dp]

def H_zero(tau, y):
    a, phi, pdot = y
    rho_tot = Omega_m / a**3 + 0.5 * pdot**2 + V(phi)
    return rho_tot - 1e-12
H_zero.terminal = True
H_zero.direction = -1

phi0 = 0.48
pdot0 = -Vp(phi0) / 3.0

sol = solve_ivp(system, [0.0, 15.0], [1.0, phi0, pdot0],
    events=H_zero, max_step=0.005,
    rtol=1e-10, atol=1e-12)

if sol.t_events[0].size > 0:
    tau_turn = sol.t_events[0][0]
    a_turn = sol.y_events[0][0][0]
    phi_turn = sol.y_events[0][0][1]
    t_turn_gyr = tau_turn * 14.5
    print(f"Turnaround at tau={tau_turn:.3f}")
    print(f"a={a_turn:.3f}, phi={phi_turn:.4f}")
    print(f"V(phi)={V(phi_turn):.4f}")
    print(f"Time: {t_turn_gyr:.1f} Gyr")
w0 = (0.5*pdot0**2 - V(phi0))/(0.5*pdot0**2 + V(phi0))
print(f"w0 = {w0:.4f}")
```

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